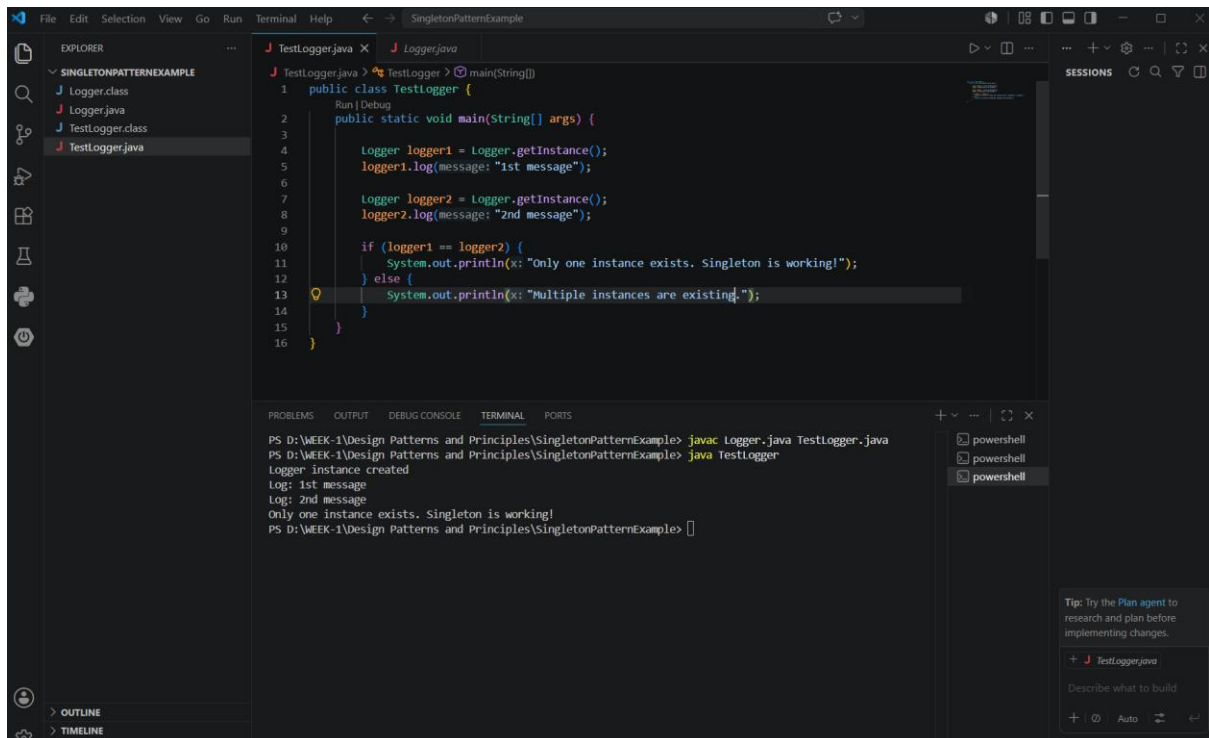


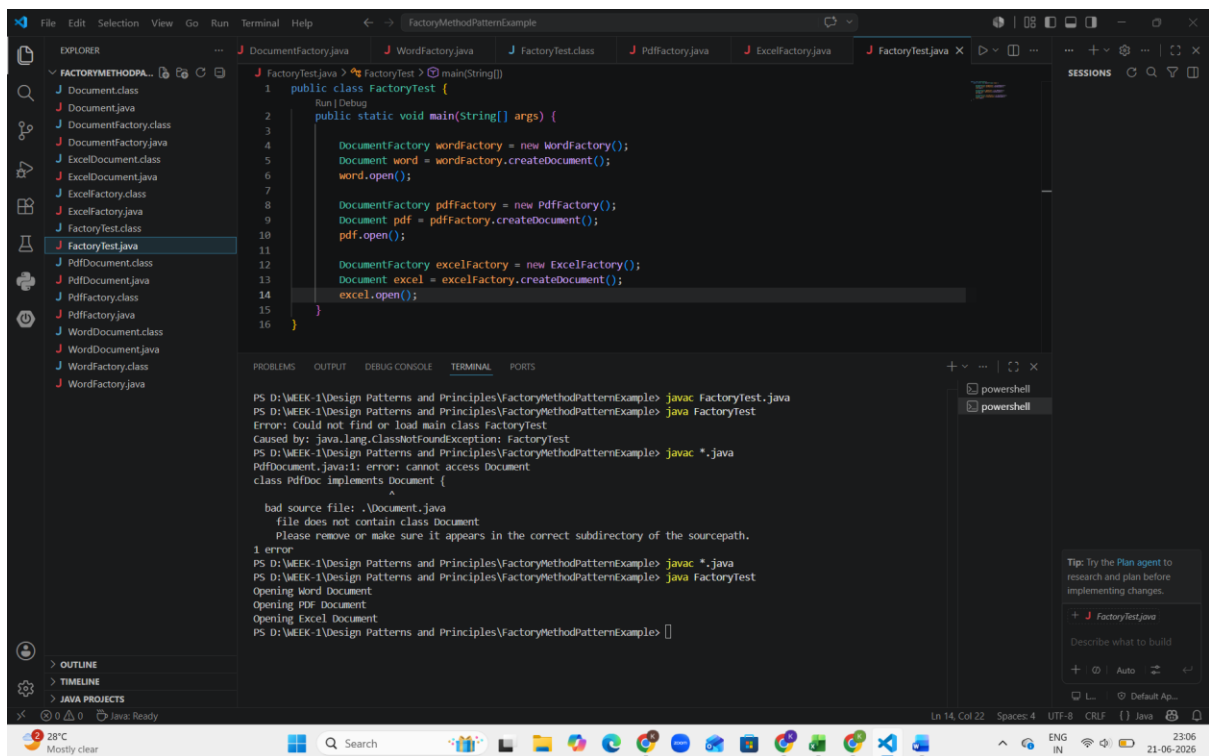
Singleton Pattern



```
1 public class TestLogger {
2     public static void main(String[] args) {
3
4         Logger logger1 = Logger.getInstance();
5         logger1.log(message: "1st message");
6
7         Logger logger2 = Logger.getInstance();
8         logger2.log(message: "2nd message");
9
10        if (logger1 == logger2) {
11            System.out.println("Only one instance exists. Singleton is working!");
12        } else {
13            System.out.println("Multiple instances are existing.");
14        }
15    }
16 }
```

PS D:\WEEK-1\Design Patterns and Principles\SingletonPatternExample> javac Logger.java TestLogger.java
PS D:\WEEK-1\Design Patterns and Principles\SingletonPatternExample> java TestLogger
Logger instance created
Log: 1st message
Log: 2nd message
Only one instance exists. Singleton is working!
PS D:\WEEK-1\Design Patterns and Principles\SingletonPatternExample> []

Factory Method Pattern



```
1 public class FactoryTest {
2     public static void main(String[] args) {
3
4         DocumentFactory wordFactory = new WordFactory();
5         Document word = wordFactory.createDocument();
6         word.open();
7
8         DocumentFactory pdfFactory = new PdfFactory();
9         Document pdf = pdfFactory.createDocument();
10        pdf.open();
11
12        DocumentFactory excelFactory = new ExcelFactory();
13        Document excel = excelFactory.createDocument();
14        excel.open();
15    }
16 }
```

PS D:\WEEK-1\Design Patterns and Principles\FactoryMethodPatternExample> javac FactoryTest.java
PS D:\WEEK-1\Design Patterns and Principles\FactoryMethodPatternExample> java FactoryTest
Error: Could not find or load main class FactoryTest
Caused by: java.lang.ClassNotFoundException: FactoryTest
PS D:\WEEK-1\Design Patterns and Principles\FactoryMethodPatternExample> javac *.java
PdfDocument.java:1: error: cannot access Document
class PdfDoc implements Document {
 A
 bad source file: .\Document.java
 file does not contain class Document
 Please remove or make sure it appears in the correct subdirectory of the sourcepath.
1 error
PS D:\WEEK-1\Design Patterns and Principles\FactoryMethodPatternExample> javac *.java
PS D:\WEEK-1\Design Patterns and Principles\FactoryMethodPatternExample> java FactoryTest
Opening Word Document
Opening Pdf Document
Opening Excel Document
PS D:\WEEK-1\Design Patterns and Principles\FactoryMethodPatternExample> []